

Charger (BQ24250) — Working Document

Digital Audio Player · Rev 3

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1. Part Identification

Field	Value
Manufacturer / MPN	Texas Instruments — BQ24250RGET
AL-PN	#1-054
Package	24-VQFN (4 × 4)
Function	3 MHz switch-mode Li-ion charger with PowerPath

2. Design Requirements

Requirement	Value
Input	USB-C, 5 V
Battery	1S Li-Po, 3.0 – 4.2 V, 1850 mAh
Charge current	500 mA (0.27C)
Input current limit	1 A
VIN_DPM	4.488 V
Switching frequency	3 MHz (2.7 – 3.3 MHz)
Charge temperature window	TS network, 0 – 60 °C
System peak (S4)	385 mA at 3.7 V (Power Budget Rev 4)

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	CE	10 kΩ pull-up to +3V3
2	EN1	10 kΩ pull-up to +3V3
3	EN2	10 kΩ pull-down to GND
4	AGND	Ground
5	SDA	10 kΩ pull-up to +3V3
6	SCL	10 kΩ pull-up to +3V3
7	STAT	Open drain — no pull-up
8	INT	10 kΩ pull-up to +3V3
9	TS	R2 from LDO, R3 + NTC to AGND
10	ISSET	RISET to AGND
11, 12	BAT	BAT+, CBAT
13, 14	SYS	SYS, CSYS bank
15, 16	PGND	GND

Pin	Name	Connection
17, 18	SW	L, CBOOT
19	IN	VUSB (filtered), CIN, VDPM divider
20	PMID	CPMID to GND
21	BOOT	CBOOT to SW
22	ILIM	RILIM to AGND
23	VDPM	Divider tap
24	LDO	CLDO, top of TS network
25	Thermal pad	GND

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
L	1.0 μ H $\pm$ 20%	2 A I_{sat} , 96 m Ω , 0505	TDK TFM141206BLE-1R0MTCA	#1-063
CBAT, CLDO, CPMID	1 μ F	16 V X6S 0402	Murata GRM155C81C105KE11D	#1-058
CBOOT	33 nF	16 V X7R 0402	Murata GRM155R71C333KA01D	#1-059
CIN (C3)	2.2 μ F	25 V X7R 0603	Murata GRM188Z71E225KE43D	#1-070
CSYS1, CSYS2	22 μ F	25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
CSYS3	0.1 μ F	50 V X7R 0402	Murata GRM155R71H104KE14D	#1-048
RISET	500 Ω 0.1%	0603	YAGEO RT0603BRC07500RL	#1-064
RILIM	270 Ω 1%	0603	YAGEO RC0603FR-07270RL	#1-065
R_DPM1 / R_DPM2	274 k Ω / 100 k Ω 1%	0603	Generic	—
R2 (TS)	5 k Ω 1%	0402	YAGEO RC0402FR-075KL	#1-060
R3 (TS)	9.88 k Ω 0.5%	0402	YAGEO RT0402DRD079K88L	#1-061
NTC	10 k Ω , β 4000 K	On cell body	—	—
F1	120 Ω @ 100 MHz	3 A, 0805	Würth 742792023	#1-068
C1, C2 (filter)	4.7 μ F	16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
R1 (filter damping)	0.5 Ω 1%	0402	YAGEO RL0402FR-070R5L	#1-072
D1	SMAJ7.0A TVS	SMA	Diodes Inc. SMAJ7.0A-13-F	#1-049
J1	USB-C receptacle	USB 2.0	GCT USB4085-GF-A	#1-069
J	Battery connector, 3-pos	5.08 mm TH	Phoenix Contact 1786417	#1-062
R_CC1, R_CC2	5.1 k Ω	0402	Generic	—
Pull-ups / pull-down	10 k Ω ($\times$ 5 / $\times$ 1)	0603	Generic	—

5. Equations and Results

Inductor ripple (V_{IN} 5 V, V_{BAT} 3.0 V, f_s 2.7 MHz, L 1.0 μ H)

$$\Delta I_L = \frac{V_{BAT}(V_{IN} - V_{BAT})}{V_{IN} f_s L} = 0.444 \text{ A}$$

Inductor peak current (Load = $I_{SYS} + I_{CHG}$)

$$I_{PEAK} = I_{Load} + \frac{\Delta I_L}{2}$$

Case	I_avg	I_PEAK
Charging, idle	0.500 A	0.722 A
Charging + S2 playback	0.637 A	0.859 A
Charging + S4 peak	0.885 A	1.107 A (vs 2 A I_sat)

Charge current

$$R_{ISET} = \frac{250}{I_{FC}} = \frac{250}{0.5} = 500 \Omega$$

Input current limit

$$R_{ILIM} = \frac{270}{I_{IC}} = \frac{270}{1.0} = 270 \Omega$$

Input DPM threshold

$$V_{IN_DPM} = 1.2 \frac{R_1 + R_2}{R_2} = 1.2 \times \frac{374}{100} = 4.488 \text{ V}$$

TS thresholds (ratiometric to V_LDO; 10 kΩ β 4000 K NTC)

Threshold	V_TS / V_LDO	Temperature
Cold	60%	0 °C
Cool	56.5%	≈ 13 °C
Warm	38.3%	≈ 47 °C
Hot	30%	60 °C

6. Important Values

Item	Value
Input filter insertion loss	−83 dB at 3 MHz
Filter output impedance peak	0.46 Ω with damping (46 Ω without); limit 6.25 Ω
SYS ripple (sim)	2.4 mV pp

Item	Value
Inductor DCR loss (charging + playback)	≈ 39 mW
Inductor peak, worst case	1.107 A (55% of 2 A I_{sat})
Battery FET $R_{ON}(BAT-SYS)$	30 m Ω typ / 40 m Ω max
IC current limit	2.6 A min

7. Open Problems

#	Problem
1	RISSET shows 50 Ω on schematic — library Value field must read 500 Ω
2	STAT has no pull-up
3	CE pulled up to +3V3 — charging disabled until firmware drives it low; defeats standalone fallback
4	VDPM upper resistor 270 k Ω on schematic vs 274 k Ω calculated
5	USB-C D+/D– (DP1/DP2, DN1/DN2) not connected — needed for USB MSC
6	AGND / GND — confirm single net or deliberate tie
7	I2C may only work while USB is present (powered from IN) — verify before driver
8	NTC return shares battery return on J pin 3
9	Battery connector height vs 13.2 mm enclosure
10	Cell limits charging to 0.2C (380 mA) at 0 - 15 °C and forbids charging above 50 °C; charger runs 500 mA until ≈ 13 °C and charges to 60 °C
11	Filter R1 pulse rating during hot plug
12	Re-simulate SYS ripple with final 2×22 μ F + 0.1 μ F bank

8. Revision Log

Rev	Date	Change
0	2026-09-19	Initial issue: part selection, sizing, calculations, TS network, input filter and SYS ripple simulations.
1	2026-09-19	Reviewed against schematic BQ-CHARGER-V1; pin connections and schematic findings added.
2	2026-09-21	Low noise and small footprint constraints added; cell charge-temperature mismatch added; condensed to essentials.
3	2026-09-22	S4 system peak 462 $\rightarrow$ 385 mA (0.6 V RMS max output): inductor peak 1.184 $\rightarrow$ 1.107 A.